

# Migration Validation & Risk Audit

Client: TestClient  
Migration: TestMig  
Date: 2026-02-21  
Auditor: Independent Migration Audit

## Executive Summary

Final Verdict: GO

|                        |      |
|------------------------|------|
| Data Volume Checks     | PASS |
| Aggregate Checks       | PASS |
| Mapping Checks         | PASS |
| Data Constraint Checks | PASS |
| Relationship Checks    | PASS |

## Data Volume Checks

| Check Name                | Status | Details                                                                          |
|---------------------------|--------|----------------------------------------------------------------------------------|
| Volume Check: users       | PASS   | Row counts match exactly for table 'users'. Source: 50, Target: 50 rows.         |
| Volume Check: orders      | PASS   | Row counts match exactly for table 'orders'. Source: 80, Target: 80 rows.        |
| Volume Check: order_items | PASS   | Row counts match exactly for table 'order_items'. Source: 120, Target: 120 rows. |

Volume Check: products

PASS

Row counts match exactly for table 'products'. Source: 30, Target: 30 rows.

## Aggregate Checks

| Check Name                         | Status | Details                                                                                                                                                                 |
|------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sum Check: users - age             | PASS   | Sum matches exactly for column 'age' in table 'users'. Source and Target both have sum 1959.<br><small>{'pct_difference': np.float64(0.0)}</small>                      |
| Average Check: users - age         | PASS   | Average matches exactly for column 'age' in table 'users'. Source and Target both have average 39.18.<br><small>{'pct_difference': np.float64(0.0)}</small>             |
| Max Check: users - age             | PASS   | Max matches exactly for column 'age' in table 'users'. Source and Target both have max 55.<br><small>{'pct_difference': np.float64(0.0)}</small>                        |
| Min Check: users - age             | PASS   | Min matches exactly for column 'age' in table 'users'. Source and Target both have min 25.<br><small>{'pct_difference': np.float64(0.0)}</small>                        |
| Variance Check: users - age        | PASS   | Variance matches exactly for column 'age' in table 'users'. Source and Target both have variance 96.885306122449.<br><small>{'pct_difference': np.float64(0.0)}</small> |
| Sum Check: users - tenure_days     | PASS   | Sum matches exactly for column 'tenure_days' in table 'users'. Source and Target both have sum 53368.<br><small>{'pct_difference': np.float64(0.0)}</small>             |
| Average Check: users - tenure_days | PASS   | Average matches exactly for column 'tenure_days' in table 'users'. Source and Target both have average 1067.36.<br><small>{'pct_difference': np.float64(0.0)}</small>   |

|                                     |      |                                                                                                                                                                     |
|-------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Max Check: users - tenure_days      | PASS | Max matches exactly for column 'tenure_days' in table 'users'. Source and Target both have max 1972.<br>{'pct_difference': np.float64(0.0)}                         |
| Min Check: users - tenure_days      | PASS | Min matches exactly for column 'tenure_days' in table 'users'. Source and Target both have min 166.<br>{'pct_difference': np.float64(0.0)}                          |
| Variance Check: users - tenure_days | PASS | Variance matches exactly for column 'tenure_days' in table 'users'. Source and Target both have variance 309923.37795918365.<br>{'pct_difference': np.float64(0.0)} |
| Sum Check: orders - amount          | PASS | Sum matches exactly for column 'amount' in table 'orders'. Source and Target both have sum 83775.83000000002.<br>{'pct_difference': np.float64(0.0)}                |
| Average Check: orders - amount      | PASS | Average matches exactly for column 'amount' in table 'orders'. Source and Target both have average 1047.1978750000003.<br>{'pct_difference': np.float64(0.0)}       |
| Max Check: orders - amount          | PASS | Max matches exactly for column 'amount' in table 'orders'. Source and Target both have max 1992.23.<br>{'pct_difference': np.float64(0.0)}                          |
| Min Check: orders - amount          | PASS | Min matches exactly for column 'amount' in table 'orders'. Source and Target both have min 57.09.<br>{'pct_difference': np.float64(0.0)}                            |
| Variance Check: orders - amount     | PASS | Variance matches exactly for column 'amount' in table 'orders'. Source and Target both have variance 278405.6154928956.<br>{'pct_difference': np.float64(0.0)}      |
| Sum Check: orders - quantity        | PASS | Sum matches exactly for column 'quantity' in table 'orders'. Source and Target both have sum 449.<br>{'pct_difference': np.float64(0.0)}                            |

|                                     |      |                                                                                                                                                                                  |
|-------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Average Check: orders - quantity    | PASS | Average matches exactly for column 'quantity' in table 'orders'. Source and Target both have average 5.6125.<br><code>{'pct_difference': np.float64(0.0)}</code>                 |
| Max Check: orders - quantity        | PASS | Max matches exactly for column 'quantity' in table 'orders'. Source and Target both have max 10.<br><code>{'pct_difference': np.float64(0.0)}</code>                             |
| Min Check: orders - quantity        | PASS | Min matches exactly for column 'quantity' in table 'orders'. Source and Target both have min 1.<br><code>{'pct_difference': np.float64(0.0)}</code>                              |
| Variance Check: orders - quantity   | PASS | Variance matches exactly for column 'quantity' in table 'orders'. Source and Target both have variance 7.784651898734177.<br><code>{'pct_difference': np.float64(0.0)}</code>    |
| Sum Check: order_items - price      | PASS | Sum matches exactly for column 'price' in table 'order_items'. Source and Target both have sum 28328.28.<br><code>{'pct_difference': np.float64(0.0)}</code>                     |
| Average Check: order_items - price  | PASS | Average matches exactly for column 'price' in table 'order_items'. Source and Target both have average 236.069.<br><code>{'pct_difference': np.float64(0.0)}</code>              |
| Max Check: order_items - price      | PASS | Max matches exactly for column 'price' in table 'order_items'. Source and Target both have max 493.89.<br><code>{'pct_difference': np.float64(0.0)}</code>                       |
| Min Check: order_items - price      | PASS | Min matches exactly for column 'price' in table 'order_items'. Source and Target both have min 6.32.<br><code>{'pct_difference': np.float64(0.0)}</code>                         |
| Variance Check: order_items - price | PASS | Variance matches exactly for column 'price' in table 'order_items'. Source and Target both have variance 20123.940056134452.<br><code>{'pct_difference': np.float64(0.0)}</code> |

|                                        |      |                                                                                                                                                                        |
|----------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sum Check: order_items - quantity      | PASS | Sum matches exactly for column 'quantity' in table 'order_items'. Source and Target both have sum 361.<br>{'pct_difference': np.float64(0.0)}                          |
| Average Check: order_items - quantity  | PASS | Average matches exactly for column 'quantity' in table 'order_items'. Source and Target both have average 3.0083333333333333.<br>{'pct_difference': np.float64(0.0)}   |
| Max Check: order_items - quantity      | PASS | Max matches exactly for column 'quantity' in table 'order_items'. Source and Target both have max 5.<br>{'pct_difference': np.float64(0.0)}                            |
| Min Check: order_items - quantity      | PASS | Min matches exactly for column 'quantity' in table 'order_items'. Source and Target both have min 1.<br>{'pct_difference': np.float64(0.0)}                            |
| Variance Check: order_items - quantity | PASS | Variance matches exactly for column 'quantity' in table 'order_items'. Source and Target both have variance 1.8570728291316525.<br>{'pct_difference': np.float64(0.0)} |
| Sum Check: products - price            | PASS | Sum matches exactly for column 'price' in table 'products'. Source and Target both have sum 12119.300000000001.<br>{'pct_difference': np.float64(0.0)}                 |
| Average Check: products - price        | PASS | Average matches exactly for column 'price' in table 'products'. Source and Target both have average 403.9766666666667.<br>{'pct_difference': np.float64(0.0)}          |
| Max Check: products - price            | PASS | Max matches exactly for column 'price' in table 'products'. Source and Target both have max 968.18.<br>{'pct_difference': np.float64(0.0)}                             |
| Min Check: products - price            | PASS | Min matches exactly for column 'price' in table 'products'. Source and Target both have min 55.28.<br>{'pct_difference': np.float64(0.0)}                              |

|                                           |      |                                                                                                                                                                                       |
|-------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variance Check: products - price          | PASS | Variance matches exactly for column 'price' in table 'products'. Source and Target both have variance 97809.61317471262.<br><code>{'pct_difference': np.float64(0.0)}</code>          |
| Sum Check: products - stock_quantity      | PASS | Sum matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have sum 7410.<br><code>{'pct_difference': np.float64(0.0)}</code>                        |
| Average Check: products - stock_quantity  | PASS | Average matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have average 247.0.<br><code>{'pct_difference': np.float64(0.0)}</code>               |
| Max Check: products - stock_quantity      | PASS | Max matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have max 496.<br><code>{'pct_difference': np.float64(0.0)}</code>                         |
| Min Check: products - stock_quantity      | PASS | Min matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have min 17.<br><code>{'pct_difference': np.float64(0.0)}</code>                          |
| Variance Check: products - stock_quantity | PASS | Variance matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have variance 24260.96551724138.<br><code>{'pct_difference': np.float64(0.0)}</code> |

## Mapping Checks

| Check Name            | Status | Details                                    |
|-----------------------|--------|--------------------------------------------|
| Mapping Check: users  | PASS   | All mappings are valid for table 'users'.  |
| Mapping Check: users  | PASS   | All mappings are valid for table 'users'.  |
| Mapping Check: orders | PASS   | All mappings are valid for table 'orders'. |

|                            |      |                                                 |
|----------------------------|------|-------------------------------------------------|
| Mapping Check: orders      | PASS | All mappings are valid for table 'orders'.      |
| Mapping Check: order_items | PASS | All mappings are valid for table 'order_items'. |
| Mapping Check: products    | PASS | All mappings are valid for table 'products'.    |
| Mapping Check: products    | PASS | All mappings are valid for table 'products'.    |

## Data Constraint Checks

| Check Name                                   | Status | Details                                                |
|----------------------------------------------|--------|--------------------------------------------------------|
| Data Constraints Check: users.first_name     | PASS   | All data constraints are satisfied for table 'users'.  |
| Data Constraints Check: users.last_name      | PASS   | All data constraints are satisfied for table 'users'.  |
| Data Constraints Check: users.email          | PASS   | All data constraints are satisfied for table 'users'.  |
| Data Constraints Check: users.date_of_birth  | PASS   | All data constraints are satisfied for table 'users'.  |
| Data Constraints Check: users.date_onboarded | PASS   | All data constraints are satisfied for table 'users'.  |
| Data Constraints Check: orders.order_date    | PASS   | All data constraints are satisfied for table 'orders'. |
| Data Constraints Check: orders.amount        | PASS   | All data constraints are satisfied for table 'orders'. |
| Data Constraints Check: orders.user_id       | PASS   | All data constraints are satisfied for table 'orders'. |

|                                                    |      |                                                             |
|----------------------------------------------------|------|-------------------------------------------------------------|
| Data Constraints Check:<br>order_items.order_id    | PASS | All data constraints are satisfied for table 'order_items'. |
| Data Constraints Check:<br>order_items.product_id  | PASS | All data constraints are satisfied for table 'order_items'. |
| Data Constraints Check:<br>order_items.quantity    | PASS | All data constraints are satisfied for table 'order_items'. |
| Data Constraints Check:<br>order_items.price       | PASS | All data constraints are satisfied for table 'order_items'. |
| Data Constraints Check:<br>products.product_name   | PASS | All data constraints are satisfied for table 'products'.    |
| Data Constraints Check:<br>products.price          | PASS | All data constraints are satisfied for table 'products'.    |
| Data Constraints Check:<br>products.stock_quantity | PASS | All data constraints are satisfied for table 'products'.    |

## Relationship Checks

| Check Name                        | Status | Details                                                                                                         |
|-----------------------------------|--------|-----------------------------------------------------------------------------------------------------------------|
| Foreign Key Check: orders         | PASS   | All foreign key values in 'user_id' of child table 'orders' have matching primary keys in parent table.         |
| Foreign Key Check:<br>order_items | PASS   | All foreign key values in 'order_id' of child table 'order_items' have matching primary keys in parent table.   |
| Foreign Key Check:<br>order_items | PASS   | All foreign key values in 'product_id' of child table 'order_items' have matching primary keys in parent table. |

## Final Deployability Verdict

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GO